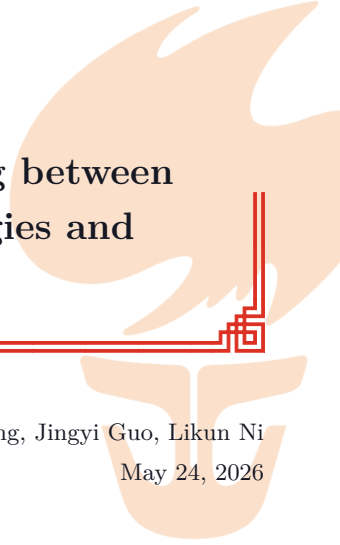


A Study on Precise Matching between Securities Investment Strategies and Client Trading Profiles



Yuhan Lei, Luoyuan Wang, Jingyi Guo, Likun Ni
May 24, 2026



Project Overview

Data and Feature Construction

Multivariate Analysis and Clustering

Client-Strategy Matching Results

Conclusion and Future Work



Project Overview



Motivation: Why Matching Matters



Quantitative strategies differ in:

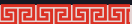
- stock selection logic
- trading frequency
- holding horizon
- concentration
- volatility and drawdown control

Clients also differ in:

- risk preference
- trading behavior
- investment style

Therefore, a useful recommendation system should satisfy: the style is close to the customer, and the risk return is better than the customer's historical performance.





This project answers three questions:

- Can we extract meaningful features from strategy and client transaction records?
- Do clients and strategies form stable groups in a multivariate feature space?
- Can we build a matching model considering both style similarity and performance improvement?

Project Overview

Goal: Build an interpretable and data-driven framework for matching quantitative stock strategies with client trading profiles.

Data:

- 20 quantitative strategies, 3 real clients, and 25 synthetic clients.
- 48 entities in total.
- 18 features covering trading behavior, portfolio structure, and risk-return characteristics.

Methods:

- PCA summarizes the main variation directions.
- K-means, hierarchical clustering, and GMM examine grouping stability.
- LDA is used only for visualization.
- A two-stage matching model combines style similarity and performance advantage.

Main results:

- The first principal component explains 44.41% of the total variance, and the first five components explain 80.38% cumulatively.
- ARI: 0.8438 between K-means and hierarchical clustering; 0.9699 between K-means and GMM.
- The final recommendations are:
Client A – Shuangchuang 50 Enhancement, score 0.4810;
Client B – Medical ETF Enhancement, score 0.4341;
Client C – CSI 2000 Enhancement, score 0.5620.

Conclusion: In the recommended strategy for each of three real clients, both return and maximum have been improved, making the framework an exploratory but feasible prototype.

Data and Feature Construction



Data construction

We used 25 simulated customers to expand the method test space:

Data construction of the analytical sample

Entity Type	Number
Quantitative strategies	20
Real clients	3
Synthetic clients	25
Total entities	48

- Real clients are limited.
- Synthetic clients are used to enlarge the methodological testing space.
- The final model is exploratory, not yet a production-level system.

Feature System

We constructed a feature system across three groups: trading behavior, portfolio structure, and risk-return performance. These features capture various aspects of trading style and performance, enabling a comprehensive analysis of both strategies and clients.

Feature Group	Examples
Trading behavior	log trade frequency, average holding days, turnover rate
Portfolio structure	buy-sell ratio, number of stocks, concentration, average position ratio
Risk-return performance	average return, win rate, profit-loss ratio, maximum drawdown, volatility, Sharpe-like ratio

Important Data Assumptions

Note that there's no complete net account value curve on the customer side, so the income, withdrawal and Sharpe like indicators are approximate estimates.

- Strategy-side annual return and drawdown are mainly taken from strategy summary workbooks.
- Strategy-side trade-level data are used to estimate win rate, volatility, and related metrics.
- Client-side complete equity curves are unavailable.
- Client-side return, drawdown, and Sharpe-like measures are approximated by FIFO-matched transactions.

Synthetic Client Design

Types of synthetic clients

Type	Main Characteristics
Conservative defensive	Lower trading frequency, stable positions, lower volatility
Balanced allocation	Moderate diversification and medium frequency
High-frequency trading	Higher turnover and shorter holding period
Growth aggressive	Higher volatility for higher potential return
Concentrated holding	Fewer stocks and higher concentration

Multivariate Analysis and Clustering



All numerical variables are standardized before analysis.

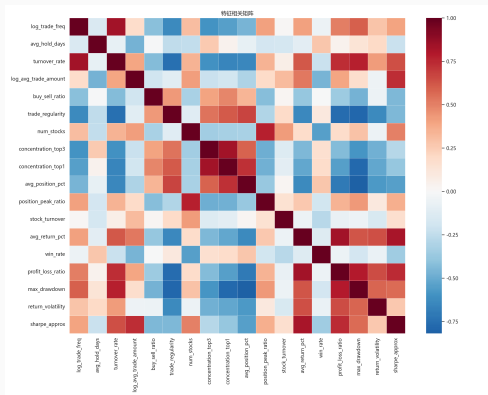
Reason:

- features have different units and scales;
- distance-based methods would otherwise be dominated by large-scale variables.

PCA is used to:

- summarize high-dimensional correlation structure;
- visualize the main variation directions;
- support interpretation before clustering.

Correlation Structure



- The 18 variables show clear correlations.
- The feature system is structured but not fully redundant.
- This supports the use of PCA.

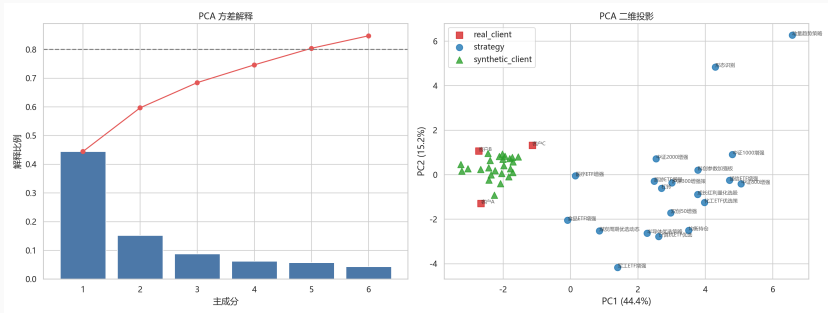
PCA Explained Variance

Explained variance ratios of the first five principal components

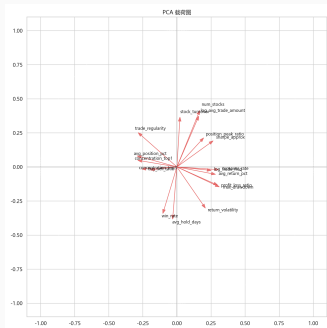
PC	Explained Variance	Cumulative
PC1	44.41%	44.41%
PC2	15.23%	59.64%
PC3	8.78%	68.42%
PC4	6.19%	74.61%
PC5	5.77%	80.38%

The first principal component accounted for 44.41%, and the first five principal components accounted for 80.38%. This shows that the feature space has a strong common structure, but not a single dimension can explain all the information.

PCA Projection



- Real and synthetic clients are relatively compact.
- Strategies are more dispersed.
- PC1 mainly reflects scale, activity, and risk-return profile.
- PC2 helps separate volatility and concentration styles.



- Features pointing in similar directions are positively associated.
- Features pointing in opposite directions are negatively associated.
- Loading plot helps interpret the economic meaning of principal components.

We compare three clustering methods:

Clustering methods used in the analysis

Method	Role
K-means	Main partitioning method
Hierarchical clustering	Robustness comparison
Gaussian mixture model	Probabilistic alternative

If the groups obtained by different methods are highly consistent, then the result is unlikely to be just an accidental product of an algorithm.

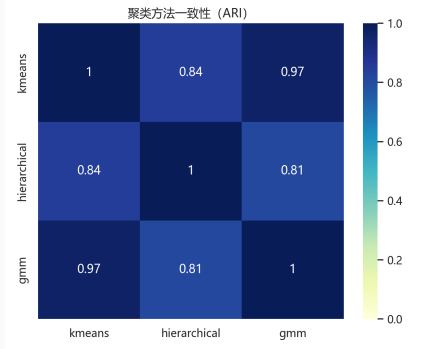
Choosing the Number of Clusters

Clustering evaluation for different values of K

K	Silhouette	Davies-Bouldin
2	0.4159	1.1418
3	0.4203	1.0752
4	0.4182	0.7337
5	0.3530	1.0175
6	0.3489	1.1852

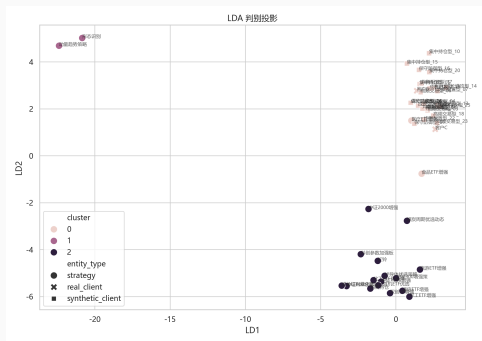
- Silhouette is highest at $K = 3$.
- Davies-Bouldin is best at $K = 4$.
- $K = 3$ is retained because it is numerically competitive and easier to interpret.

Clustering Robustness



- ARI between K-means and hierarchical clustering: 0.8438.
- ARI between K-means and GMM: 0.9699.
- High agreement suggests stable grouping structure.

LDA Visualization



- LDA is used only for visualization.
- It is not an independent validation method.
- Clients are more concentrated.
- Some strategies are far from the central cloud.

Client-Strategy Matching Results



Two-Stage Matching Model

Stage 1: Style similarity

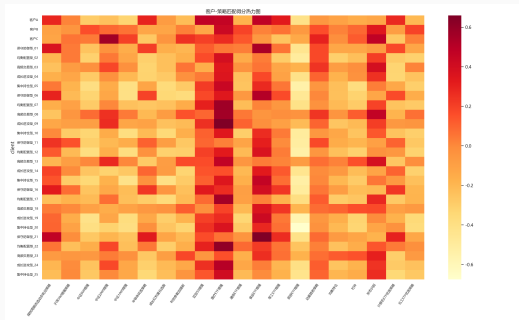
style score = $0.45 \times \text{behavior similarity} + 0.55 \times \text{holding similarity}$.

Stage 2: Performance advantage

- return comparison
- Sharpe-like ratio comparison
- maximum drawdown comparison
- volatility comparison

final score = $0.7 \times \text{style score} + 0.3 \times \text{performance score}$.

Matching Score Heatmap



- Rows: three real clients; Columns: candidate strategies; Color: final matching score.
- Different clients receive different high-score strategies.

Not all customers push the same high-yield strategy, which shows that the *style component* is really distinguishing customers.

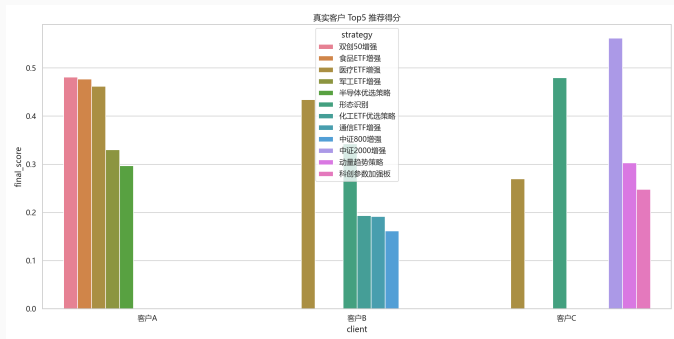
Top Recommendations

Top recommended strategy for each real client

Client	Top Recommended Strategy	Final Score	Return Improvement	Drawdown Improvement
Client A	Shuangchuang 50 Enhancement	0.4810	1.1015	0.2580
Client B	Medical ETF Enhancement	0.4341	0.2115	0.8010
Client C	CSI 2000 Enhancement	0.5620	0.6706	0.3372

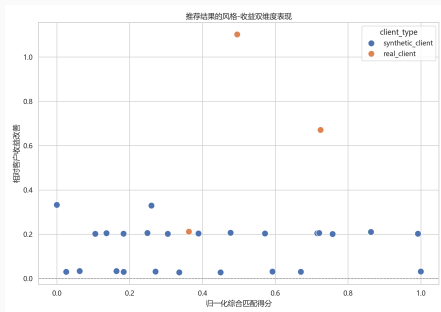
Three real customers get different top strategies. More importantly, the three top strategies have improved both return and maximum drawdown.

Top-5 Recommendations and Trade-off



- Top five recommended strategies differ across clients.
- The model gives client-specific recommendation lists.

Top-5 Recommendations and Trade-off



This figure further demonstrates the relationship between style matching and performance improvement:

some strategy styles are more similar, and some strategy improvements are more obvious.

In actual business, it may be necessary to weigh the two.

Kruskal-Wallis test results for selected features

Feature	p -value
win rate	0.0298
position peak ratio	0.0322
return volatility	0.0324
number of stocks	0.0369
approximate Sharpe ratio	0.0894

Kruskal-Wallis test cannot prove causality, nor can it prove that clustering is a real market structure. But it shows that some economically meaningful variables do differ among different clusters, such as *win rate*, *position peak ratio*, *return volatility* and *number of stocks*.

Conclusion and Future Work



Summary: Main Findings

1. The feature system captures meaningful differences among clients and strategies.
2. $K = 3$ clustering gives a stable and interpretable grouping structure.
3. Clustering results are consistent across K-means, hierarchical clustering, and GMM.
4. The two-stage matching model produces client-specific recommendations.
5. For all three real clients, the top strategy improves both return and maximum drawdown.

Limitations

1. Only three real client accounts are available.
2. Synthetic clients are used for methodological expansion, not for proving real investor distribution.
3. Client-side performance indicators are approximate.
4. Complete daily account equity curves are unavailable.
5. Matching weights are rule-based and not learned from real client acceptance or future performance.

1. Collect more real client accounts.
2. Use complete daily account-level equity curves.
3. Add richer industry preference and sector exposure variables.
4. Learn matching weights from historical recommendation outcomes.
5. Evaluate recommendations using out-of-sample future performance.

Conclusion

- We build an interpretable workflow for matching clients and quantitative strategies.
- The workflow combines feature engineering, PCA, clustering, LDA visualization, and two-stage matching.
- The results show stable grouping patterns and client-specific strategy recommendations.
- The model should be interpreted as a feasible prototype, not a final deployment system.

Final takeaway

Overall, our project shows that transaction-based client profiles and strategy characteristics can be integrated into a common multivariate framework, making client-specific and interpretable strategy recommendation possible.

Thanks

